

# Koduru Suchith

Computer Science Engineering Graduate, Indian Institute of Technology Delhi '25

✉ suchithkoduru@gmail.com | 💻 koduru-suchith | 🌐 suchith83 | 🌐 suchith.space | 📞 +91-8465073250

## PROFESSIONAL SUMMARY

AI & Full-Stack Engineer with **1+ year of experience** building production **LLM agents**, **RAG pipelines**, and **real-time voice AI** at a fast-paced startup. Strong in **FastAPI**, **MCP servers**, and **streaming** (WebSockets, SSE), with hands-on LLM orchestration (**AgentFlow**, **LangGraph**), vector search (**Qdrant**), and edge inference (**Jetson**, **Raspberry Pi 5**). Open-source contributor to **AgentFlow**. B.Tech, IIT Delhi.

## EDUCATION

B.Tech, Computer Science & Engineering | Indian Institute of Technology Delhi August, 2021 – May, 2025

## WORK EXPERIENCE

AI Software Engineer – 10xScale.ai | Hyderabad, India (January'26 – June'26)

**Domain-Specialized Conversational Agent:**

- Built a domain-specialized medical agent on **AgentFlow** (LangGraph-like **StateGraph** + **ToolNode**), with per-skill playbooks (symptom triage, appointment booking, medication guidance) loaded at runtime, served via **FastAPI** with **SSE**-streamed responses to frontend client.
- Persisted conversation state with a checkpointer for short-term context and **Qdrant** for long-term cross-session memory.
- Authored custom transactional tools (doctor availability lookup, appointment booking, structured DB queries) so the agent takes real actions, not just chat.

**Voice-Enabled Edge AI Application (Raspberry Pi 5):**

- Built a voice-driven medical-kiosk assistant on **Raspberry Pi 5 (PySide6)** with an async **VAD** → **STT** → **Agent** → **TTS** pipeline and a **FastAPI** backend; supports 11 Indic languages with per-turn auto-detection.
- Owned the **end-to-end software development and integration**, delivering and connecting the **backend APIs**, **AI agent service**, **web application**, and **Raspberry Pi 5 desktop application** into a unified production system.
- Integrated **WhatsApp** via **Twilio**: inbound webhooks extract PDFs/images using **Gemini Vision** into agent context; outbounds consultation summaries are generated via **Gemini 2.5 Flash**.
- Reduced barge-in latency by **~70%** (1000 ms → 300 ms) by flushing the OS audio buffer on stop-TTS and chunking TTS at the first sentence.

**Enterprise AI Interview Platform (Voice Agent):**

- Owned the **AI interviewer agent logic** for **CV**, **DSA-coding**, **system-design**, and **behavioral rounds**, including dynamic question selection, round progression, evaluation flow, and persistent interview state using an MCP-based question bank.
- Built **backend evaluation APIs** to execute submitted coding snippets against test cases and evaluate candidate performance, enabling automated assessment of coding rounds.
- Developed an evaluation pipeline that aggregates **per-round evaluations and the complete interview transcript** to generate a structured **final candidate report**.
- Engineered the **real-time voice loop** (**STT** → **LLM** → **TTS**) to achieve **sub-700 ms** turn latency through VAD-based endpointing, TTS pre-warming, token streaming, and barge-in support.
- Contributed to the **Electron + React desktop application**, integrating the agent and voice services through a **STDIO JSON-RPC** sidecar and enabling streaming transcripts, responses, and agent/tool events.

**General Experience:**

- Deployed services with **Docker** and configured routing, upstreams, and WebSocket passthrough in **Apache APISIX**.
- Benchmarked local LLM inference on **NVIDIA Jetson Orin Nano** (Ollama 7B, MedGemma 4B, Whisper, Qwen TTS) for edge AI viability.

AI Software Engineer Internship – 10xScale.ai | Hyderabad, India (June'25 – December'25)

**MCP Server & Database-Aware Agent:**

- Built a custom **MCP server (FastMCP)** exposing **PostgreSQL** natural-language-to-SQL and **Typesense** (keyword + embedding ANN) search tools over structured course and job data, secured with auth middleware.
- Wired the server into a **database node** of a multi-agent system that auto-discovers the MCP tools, plans tool calls, and grounds responses in real data, eliminating hallucinations on factual questions.

**AgentFlow – Open Source Contributions:**

- Contributed to open-source **AgentFlow**: shipped a **Qdrant**-backed long-term memory module and hardened the **Google GenAI** adapter against empty / None-content edge cases (with regression tests). **2 PRs merged** (#17, #19).

## KEY PROJECTS

Multi-Class Image Classification (CIFAR-100) | Machine Learning (Sep'24 – Nov'24)

- Developed a high-confidence image classification model on the **CIFAR-100** dataset, experimenting with multiple architectures and calibration methods using **PyTorch**.
- Implemented **WideResNet (WRN-28-10)** and boosted generalisation using **Sharpness-Aware Minimization (SAM)**.
- Enhanced prediction reliability through **temperature scaling** calibration, improving calibrated accuracy on high-confidence outputs.

## TECHNICAL SKILLS

|           |                                                                             |                       |                                                                                          |
|-----------|-----------------------------------------------------------------------------|-----------------------|------------------------------------------------------------------------------------------|
| Languages | Python, C++ , SQL, JavaScript/TypeScript                                    | AI / ML               | PyTorch, AgentFlow, LangGraph, LangChain, MCP / FastMCP, RAG, Gemini, SHAP, Scikit-Learn |
| Backend   | FastAPI, async Python, PostgreSQL, Qdrant, Typesense, REST, WebSockets, SSE | Frontend              | React, PySide6, Electron, Streamlit                                                      |
| DevOps    | Docker, Git, GitHub Actions, Apache APISIX, GCP / GCS                       | AI Coding As-sistants | Claude Code, Cursor                                                                      |